

Quiz (Transactions and Locks I)

Instructions: For each problem, you must perform all of the following tasks correctly in order to get credit.

1. Circle all SQL commands that block.
2. Write an X through all commands that error.
3. If a command causes a deadlock, write DEADLOCK and stop the problem.
4. Write the output of each SELECT statement.

You are allowed to use a computer and access the internet. (I strongly encourage you to refer to the postgres documentation.) BUT YOU ARE NOT ALLOWED TO CONNECT TO POSTGRES AND RUN POSTGRES COMMANDS.

Problem 1:

Session 1	Session 2
1 CREATE TABLE t (a);	1
2 CREATE TABLE u (b);	2
3 BEGIN;	3
4	4 BEGIN ISOLATION LEVEL
5	5 REPEATABLE READ;
6 INSERT INTO t VALUES (9);	6
7	7 SELECT count (*) FROM u;
8 COMMIT;	8
9	9 INSERT INTO u VALUES (9);
10 SELECT count (*) FROM u;	10
	11 SELECT count (*) FROM u;
	12 COMMIT;

Problem 2:

Session 1

```
1 CREATE TABLE t (a int);
2 BEGIN ISOLATION LEVEL
3 READ UNCOMMITTED;
4 INSERT INTO t VALUES (5);
5
6
7
8 INSERT INTO t VALUES (6);
9
10
11
12 COMMIT;
```

Session 2

```
1
2
3
4
5 BEGIN;
6 SELECT count(*) FROM t;
7 INSERT INTO t VALUES (6);
8
9 SELECT count(*) FROM t;
10 INSERT INTO t VALUES (5);
11 ABORT;
12
13 SELECT count(*) FROM t;
```

Problem 3:

Session 1

```
1 CREATE TABLE t ( a INT );  
2 BEGIN;  
3 LOCK TABLE t IN SHARE MODE;  
4  
5 COMMIT;
```

Session 2

```
1  
2  
3  
4 VACUUM t;
```

Problem 4:

Session 1	Session 2
1 CREATE TABLE t (a INT);	1
2 INSERT INTO t VALUES (1);	2
3 INSERT INTO t VALUES (2);	3
4 INSERT INTO t VALUES (3);	4
5	5 BEGIN;
6 BEGIN;	6
7 SELECT * FROM t WHERE a=2 FOR UPDATE;	7
8	8 UPDATE t SET a=7 WHERE a=2;
9	9 COMMIT;
10 COMMIT;	